

Glucose, Insulin and Diabetes

Beta cells (TCF7L2, MTNR1B, SLC30A8, KCNJ11), insulin sensitivity (PPARG, IRS1, GCKR), adiposity (FTO, MC4R), autoimmunity (PTPN22) and 2 type diabetes scores, 1, HbA1c, glucose and fasting insulin

Sérgio F · 02-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. The **HbA1c** genetic scores are very low (6.6 and 8.4 percentiles), such as fasting glucose (19 variants, percentile 9.5). Genetically, the mean glucose tends to be below the average of the population. There is no risk allele in HNF1A, GCK, MC4R, ADIPOQ or PTPN22. Type 1 diabetes is typical to low (41.9 and 29.4 percentiles).

Unfavourable. The polygenic risk of 2 type diabetes is typical, slightly above the median (56.7, 58.6, 68.9 and 62.4 percentiles). It has risk alleles in several genes of the beta cell: TCF7L2 heterozygous (the most effective gene), SLC30A8 with 2 copies (insulin zinc carrier), KCNJ11 E23K with 2 copies, and CDKAL1, CDKN2A/B, HHEX, IGF2BP2 and ADCY5 heterozygotes. **MTNR1B** is heterozygous (insulin secretion more sensitive to melatonin, that is, late meals), and G6PC2 has 2 copies of the allele that rises glucose in fasting. There are also IRS1, KLF14 and FTO heterozygotes.

What does it mean? The profile is "beta cell with low reserve, low average glucose". Genes do not predict diabetes above average, and genetic HbA1c is favourable. But the insulin secretion alleles together indicate that the response to large glucose loads (postprandial peaks) and late meals is the weak point. The control of carbohydrates, meal time, exercise and body composition weigh more than average. The general glycemic metabolism (analyzes) is in §23; here only the genes.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Keyvariants** 20 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 20 of the chip, 0 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 1 SNPs needed conversion to the complementary strand and 4 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 11 PGS Catalog scores calculated locally with 10.450.887 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MAIS PROVÁVEL/ALTO; ≤ 20 : MENOS PROVÁVEL/BAIXO; between 20 and 80: TÍPICO.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 5888 loci, 5234 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

High percentile = higher risk (diabetes) or higher value (HbA1c, glucose, fasting insulin).

2 type diabetes

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
2 type diabetes	PGS000014 (Kera 2018, LDpred, 6.9 million variants)	6.393.464 6.917.436 (weighted coverage 89%)	56,7	TIPIC	typical
2 type diabetes	PGS002720 (Weissbrod 2022, SbayesR, 912 thousand variants)	891.114 911.809 (weighted coverage 97%)	58,6	TIPIC	typical
2 type diabetes	PGS003443 (Huerta-Chagoya 2023, PRS-CSx, 1.09 million variants)	1.069.380 1.092.496 (weighted coverage 97%)	68,9	TIPIC	typical
2 type diabetes	PGS000805 (Polfus 2021, 582 variants)	564 582 (weighted coverage 94%)	62,4	TIPIC	typical

Glycemic markers

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
HbA1c	PGS001953 (Privé 2022, 46.6 thousand variants)	44.856 46.566 (weighted coverage 96%)	6,6	BAIXA	favourable
HbA1c	PGS002697 (Weissbrod 2022, SbayesR, 989 thousand variants)	967.214 989.344 (weighted coverage 97%)	8,4	BAIXA	favourable
Fasting glucose	PGS000306 (Xie 2020, 19 variants)	19 19 (weighted coverage 100%)	9,5	BAIXA	favourable
Fasting insulin	PGS000308 (Xie 2020, 12 variants)	12 12 (weighted coverage 100%)	4,9	BAIXA	favourable
Fasting insulin	PGS005276 (Sarnowski 2025, PRS-CSx, 1.1 million variants)	1.084.190 1.116.555 (weighted coverage 95%)	74,4	TIPIC	typical

1 type diabetes

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
1 type diabetes	PGS000021 (Oram 2015, 33 variants (includes HLA))	25 25 (weighted coverage 100%)	41,9	TIPIC	typical
1 type diabetes	PGS000024 (Sharp 2019, 85 variants (GRS2, includes HLA))	49 53 (weighted coverage 86%)	29,4	TIPIC	typical

- **1 type diabetes:** The 1 type diabetes scores depend heavily on imputed HLA haplotypes; they serve to exclude, not to predict.

4. Key variants by topic

Variants with replicated effect in 2 type diabetes and glycemic traits. Each one has a small effect (OR ~1.1–1.4 per allele).

Beta cells and insulin secretion

Most of the 2 type diabetes genes act on the pancreatic beta cell: how much insulin is made, when released and how the cell survives stress (TCF7L2, SLC30A8, KCNJ11, CDKAL1, CDKN2A/B, HHEX, IGF2BP2, ADCY5, HNF1A).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
TCF7L2	rs7903146		C/T	chip	T (1 copy/s)	unfavourable (light)	The T allele reduces insulin secretion and response to incretins (GLP-1). It is the largest common locus of diabetes type 2 (OR ~1.4 per allele)	Strong
MTNR1B	rs10830963		C/G	chip	G (1 copy/s)	unfavourable (light)	Melatonin receptor in the beta cell: G allele stops insulin secretion, especially at night (late meals)	Strong
SLC30A8	rs13266634	R325W	C/C	chip	C (2 copy/s)	unfavourable	R325W: C allele (Arg325) gives less zinc in insulin granules; T is protective	Moderate
KCNJ11	rs5219	E23K	T/T	chip	T (2 copy/s)	unfavourable	E23K: ATP sensitive potassium channel; T allele reduces insulin secretion	Moderate
CDKAL1	rs7756992		A/G	chip	G (1 copy/s)	unfavourable (light)	Proinsulin processing	Moderate
CDKN2A/B (9p21)	rs10811661		T/C	chip	T (1 copy/s)	unfavourable (light)	Mass and regeneration of beta cells (p16/p15)	Moderate
HHEX/IDE	rs1111875		C/T	chip	C (1 copy/s)	unfavourable (light)	Pancreatic development and insulin degradation	Moderate
IGF2BP2	rs4402960		G/T	chip	T (1 copy/s)	unfavourable (light)	Insulin secretion	Moderate
ADCY5	rs11708067		A/G	chip	A (1 copy/s)	unfavourable (light)	Cyclic AMP signal in beta cell (glucose 2 h ↑)	Moderate
HNF1A	rs1169288	I27L	A/A	chip	C (0 copy/s)	favourable	I27L: transcription factor of the beta cell (gene MODY3); common variant of small effect	Weak

Fasting glucose and glucose sensor

G6PC2, GCK and GCKR define the point of regulation of fasting glucose. MTNR1B binds melatonin to insulin secretion, which makes meal time relevant.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
GCK	rs1799884		C/C	chip	A (0 copy/s)	favourable	Glucokinase –30G>A: allele A rises the glucose threshold (fasting glucose ↑)	Moderate
G6PC2	rs560887		C/C	chip	C (2 copy/s)	unfavourable	The C allele rises fasting glucose (effect at the point of regulation, not at the risk of diabetes)	Moderate
GCKR	rs1260326	P446L	T/C	chip	T (1 copy/s)	mixed	P446L: the T allele lowers fasting glucose and insulin, but rises triglycerides and hepatic fat	Moderate

Insulin sensitivity and adiposity

Insulin sensitivity depends on muscle and adipose tissue (IRS1, KLF14, PPARG, ADIPOQ) and adiposity (FTO, MC4R).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PPARG	rs1801282	Pro12Ala	C/C	chip	G (0 copy/s)	neutral	Pro12Ala: G allele (Ala12) improves insulin sensitivity and protects from diabetes (OR ~0.85)	Strong
IRS1	rs2943641		T/C	chip	C (1 copy/s)	unfavourable (light)	The C allele is associated with more insulin resistance and less subcutaneous fat (metabolic adiposity)	Moderate
KLF14	rs4731702		C/T	chip	C (1 copy/s)	unfavourable (light)	Maternal regulator of adipose tissue: C allele gives greater adipocytes and more insulin resistance	Moderate
FTO	rs9939609		T/A	chip	A (1 copy/s)	unfavourable (light)	Allele A increases appetite and adiposity (it is the effect of adiposity that weighs most on diabetes); exercise attenuates it ~30%	Strong
MC4R	rs17782313		T/T	chip	C (0 copy/s)	favourable	Hypothalamic satiety: C allele increases adiposity	Moderate
ADIPOQ	rs266729		C/C	chip	G (0 copy/s)	favourable	Adiponectin promoter: G allele is associated with lower adiponectin	Weak

Autoimmunity (1 type diabetes)

Autoimmunity genes only serve to exclude 1 type diabetes (HLA is imputed).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PTPN22	rs2476601	R620W	G/G	chip	A (0 copy/s)	favourable	R620W: allele A increases the risk of autoimmunity (1 type diabetes, thyroiditis, rheumatoid arthritis)	Strong

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of each glycemic trait, with alleles oriented towards higher risk or value compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
2 type diabetes	3661 (3318)	3224 de 6636	3304	49,6	typical
Fasting glucose	585 (499)	477 de 998	494	15,2	favourable
HbA1c	1557 (1336)	1332 de 2672	1331	37,9	typical
Fasting insulin/resistance	85 (81)	83 de 162	80	64,3	typical

6. Favourable

- **Genetically low HbA1c:** percentiles 6.6 and 8.4 (scores of 46.6 thousand and 989 thousand variants). The HbA1c loci load is typical (37.9). **Moderate.**
- **Low fasting glucose:** 19 score variants in percentile 9.5; loci load in the 15.2. **Moderate.**
- **Fasting insulin:** the 12 variant score is in percentile 4.9 (the genomic one is in the 74.4, typical). **Weak.**
- **No risk alleles** in HNF1A, GCK (−30G>A), MC4R, ADIPOQ and PTPN22. **Moderate.**
- **1 type diabetes:** percentiles 41.9 and 29.4. Moderate (HLA is imputed).

7. Unfavourable

- **2 type diabetes: typical polygenic risk, slightly above the median** (percentis 56.7, 58.6, 68.9 and 62.4); loci load in 49.6. **Moderate.**
- **TCF7L2 rs7903146 C/T**: 1 risk allele (OR ~1.4 by allele). Less effect of incretins and less insulin secretion. **Strong.**
- **SLC30A8 R325W C/C** and **KCNJ11 E23K T/T**: 2 risk copies in each. They affect zinc storage in insulin granules and the potassium channel that triggers secretion. **Moderate.**
- **CDKAL1, CDKN2A/B, HHEX/IDE, IGF2BP2 and ADCY5**: 1 risk allele in each (beta cell). **Moderate in the set.**
- **MTNR1B rs10830963 C/G**: the melatonin receptor inhibits more insulin, so late meals tolerate worse. **Moderate.**
- **G6PC2 rs560887 C/C**: point of regulation of fasting glucose a little higher (small effect, without clear increase in diabetes). **Weak moderated.**
- **IRS1, KLF14 and FTO**: 1 allele in each (insulin sensitivity, adipose tissue and adiposity). **Weak moderated.**
- **GCKR P446L T/C**: mixed (fasting glucose ↓, triglycerides and hepatic fat ↑; see Liver report). **Moderate.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Hyperglycemia and Intolerantness to Glucose	HbA1c and glucose in low genetic fasting; several beta cell alleles (postprandial peaks)	Mixed
Insulin resistance	IRS1, KLF14 and FTO heterozygotes; PPARG without protection; ADIPOQ without risk; typical fasting insulin	Neutral to unfavourable (mild)
Metabolic Homeostasis Decline	Typical 2 polygenic diabetes (56.7–68.9 percentiles)	Neutral
Accumulation of Advanced Glycation Final Products (AGEs)	The low genetic HbA1c suggests less average glycation	Favourable
Circadian disorders	MTNR1B C/G: insulin secretion depends more on meal time	Unfavourable (mild)
Nutrient Sensing Deregulation	TCF7L2 (incretins) and KCNJ11/SLC30A8 (glucose-insulin coupling)	Unfavourable (mild)
Dysfunction of adipose tissue	KLF14 and FTO heterozygotes; MC4R and ADIPOQ favourable	Neutral
Proteostasis loss	The insulin synthesis load stresses the beta cell reticulum (CDKAL1, IGF2BP2, SLC30A8)	Unfavourable (mild)

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Early meals; light and early dinner** (≥ 3 h before bedtime). With MTNR1B C/G, the same late meal gives more glucose. *Suggestion (not applied)*.
2. **Flatten postprandial peaks**: start the meal with vegetables and protein; high fiber (the more, the better); hydrates of whole carbon and low glycemic index; avoid sugary beverages. *Suggestion (not applied)*.
3. **Walk 10-15 minutes after the main meals**: the muscle captures glucose without insulin. *Suggestion (not applied)*.
4. **2-3x/week and 2 zone** (as in the plane). Muscle mass is the largest glucose reservoir. *Suggestion (not applied)*.
5. **Body composition**: keeping visceral adiposity low (FTO, KLF14). *Suggestion (not applied)*.
6. **Zinc 20-25 mg/day total with copper (already in the plane)**: consistent with SLC30A8 C/C (zinc is required to store insulin). Do not increase the dose. *Suggestion (not applied)*.
7. **Regular and sufficient sleep** (MTNR1B; sleep deprivation reduces insulin sensitivity). If melatonin is used, keep the dose of the plan (0.3-1 mg) and take it away from the last meal ($\geq 2-3$ h). *Suggestion (not applied)*.
8. **Genetic-oriented surveillance**: HbA1c, glucose and insulin in annual fasting. To see the peaks, once: glucose curve or 2 weeks of continuous glucose monitor (the beta cell profile is shown at peaks, not at fasting). *Suggestion (not applied)*.
9. **Omega-3 (already in the plane) and zero alcohol (already fulfills)**: neutral to favourable for glucose. *Suggestion (not applied)*.

10. Limitations

- The 2 diabetes scores explain a small fraction of the risk. Lifestyle and body composition weigh more.
- The 1 type diabetes scores depend on imputed HLA haplotypes: they serve to exclude, not to predict.
- The two fasting insulin scores disagree (12 variants: low percentile; genomic: typical).
- The chip does not read rare variants of monogenic diabetes (MODY: HNF1A, HNF4A and GCK rare).
- The load of loci GWAS sums SNPs in linkage imbalance. It is descriptive and does not replace the polygenic scores.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the Imposed VCF; in the palindromics the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed.
- **Polygenic scores:** GRCh38 files from PGS Catalog (11 scores: 2 type diabetes, HbA1c, fasting glucose and insulin, 1 type diabetes). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 510^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** reference literature (Grant 2006 for TCF7L2; Sladek 2007 for SLC30A8; Lyssenko 2009 for MTNR1B; Mahajan 2018 and Suzuki 2024 for 2 type diabetes meta-analysis Dupuis 2010 for glycemic traits), GWAS Catalog and ClinVar. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Beta cells and insulin secretion	TCF7L2	rs7903146	C/T	chip	T	1	strand +	1 copy of risk allele
Beta cells and insulin secretion	MTNR1B	rs10830963	C/G	chip	G	1	strand + (palindromic; assumed +)	1 copy of risk allele
Beta cells and insulin secretion	SLC30A8	rs13266634	C/C	chip	C	2	strand +	2 risk allele copies
Beta cells and insulin secretion	KCNJ11	rs5219	T/T	chip	T	2	strand +	2 risk allele copies
Beta cells and insulin secretion	CDKAL1	rs7756992	A/G	chip	G	1	strand +	1 copy of risk allele
Beta cells and insulin secretion	CDKN2A/B (9p21)	rs10811661	T/C	chip	T	1	strand +	1 copy of risk allele
Beta cells and insulin secretion	HHEX/IDE	rs1111875	C/T	chip	C	1	strand +	1 copy of risk allele
Beta cells and insulin secretion	IGF2BP2	rs4402960	G/T	chip	T	1	strand +	1 copy of risk allele
Beta cells and insulin secretion	ADCY5	rs11708067	A/G	chip	A	1	strand +	1 copy of risk allele
Beta cells and insulin secretion	HNF1A	rs1169288	A/A	chip	C	0	strand +	without the risk allele
Fasting glucose and glucose sensor	GCK	rs1799884	C/C	chip	A	0	complement (strand)	without the risk allele
Fasting glucose and glucose sensor	G6PC2	rs560887	C/C	chip	C	2	strand +	2 risk allele copies
Fasting glucose and glucose sensor	GCKR	rs1260326	T/C	chip	T	1	strand +	C/T: light ↓ glucose; light ↑ triglycerides
Insulin sensitivity and adiposity	PPARG	rs1801282	C/C	chip	G	0	strand + (palindromic; assumed +)	without the protective allele (common genotype)
Insulin sensitivity and adiposity	IRS1	rs2943641	T/C	chip	C	1	strand +	1 copy of risk allele
Insulin sensitivity and adiposity	KLF14	rs4731702	C/T	chip	C	1	strand +	1 copy of risk allele
Insulin sensitivity and adiposity	FTO	rs9939609	T/A	chip	A	1	strand + (palindromic; assumed +)	1 copy of risk allele
Insulin sensitivity and adiposity	MC4R	rs17782313	T/T	chip	C	0	strand +	without the risk allele
Insulin sensitivity and adiposity	ADIPOQ	rs266729	C/C	chip	G	0	strand + (palindromic; assumed +)	without the risk allele
Autoimmunity (1 type diabetes)	PTPN22	rs2476601	G/G	chip	A	0	strand +	without the risk allele